Final Project: Description

Statistical Evaluation of AI · Spring 2027

Weight	40% of the course grade (100 project points)
Team	Four students. Team and track choices are due Thu Mar 4.
Final product	Slides and a five-minute video, due Sun Apr 25 at 11:59 pm, then an oral defense in class
Grading	The *Final Project: Rubric*

This description and the rubric are the only two project documents. Slides, Canvas pages, and starter notebooks point back to them and do not add requirements. Canvas has the official due dates. If anything else disagrees with these two documents, these documents win; please tell the instructor so it can be fixed.

1. The project

Your team works as a research team for a client. The client has an AI system, some data, and a decision to make. Your team designs and carries out a statistical study to inform that decision, then presents the study to the client as one clear argument.

You are not building a new AI system or chasing the highest accuracy. The AI system's outputs are provided. Your job is to work out what the evidence supports, for which population, and how sure the client can be. The question behind the whole project is:

> **What should a careful person conclude from this evidence, and how sure should they be?**

2. Four tracks

Track	Client and decision	Data	AI system	Fitted baseline
Food tagging	A marketplace: should it tag food photos automatically?	Food-101 photos	CLIP image classifier	Logistic regression on simple image features
Pet tagging	A marketplace: can it tag pet breeds, or only cats versus dogs?	Oxford-IIIT Pet photos	CLIP image classifier	Logistic regression on simple image features
Human preferences	A research team: does an LLM judge measure what people prefer?	Stanford Human Preferences pairs from Reddit	LLM judge (cached outputs)	Length-only and TF-IDF logistic regression
Instruction following	A research team: can an LLM judge check whether responses follow instructions?	IFEval-style prompts and responses	Rule checker and LLM judge (cached outputs)	Logistic regression on response and task features

Each track comes with a course subset of the data, the AI system's outputs, a starter notebook, a datasheet, and a system card. They are in the project data folder linked from the course website, which opens in Week 9. No track needs a GPU, a paid service, or a large download. You may give the project data and outputs to an AI tool, for example to get help with code or to look through cases. Check what it tells you against the data.

Every team also reports a **floor**, such as always predicting the most common class. The floor is not the baseline. The **fitted baseline** is a statistical model your team fits, understands, and can defend. Using a random forest instead of logistic regression needs the instructor's approval. Using your own data or AI system needs written approval by Fri Mar 19, and approval is rare.

3. One study, told as one story

Scientific papers and talks share a four-part structure called **IMRaD**: Introduction, Methods, Results, and Discussion. Your project uses it twice. The three milestones build the four sections as a short written document. The final slides and video tell the same story to the client in five minutes. The five-step workflow from class fits into the four sections.

Section	It answers	What goes in it	Workflow step
Introduction	Why does this matter, and what are we asking?	The client, the decision, and two to four questions, each tied to the decision. The population the client cares about and the population the benchmark samples. The quantity (estimand) each question needs.	1. Question and estimands
Methods	How did we find out?	The data and what exploratory analysis showed about it: balance, label quality, hard cases, and how the benchmark differs from the client's population. The floor, the fitted baseline, and the AI system. The evaluation design: unit, split, metric, and pairing. How uncertainty is computed. The sensitivity analysis, with the result that would change the recommendation, stated before it is run.	2. Exploratory analysis, and the design of steps 3 and 4
Results	What did we find?	Answers to the questions, in the order the Introduction asked them. The baseline and the AI system compared on the same cases, with uncertainty. Calibration, when the decision uses confidence scores. Where the errors concentrate. The sensitivity analysis result.	3. Modeling and inference; 4. Sensitivity analysis
Discussion	What should the client do?	The recommendation and the conditions on it. The largest limitations. The result that would change the recommendation, and the next data worth collecting.	5. Decision and limitations

What makes it one story

A set of correct analyses is not yet an argument. Five rules turn it into one.

1. **Every section answers to the Introduction.** The Results answer the Introduction's questions in the same order, and the Discussion makes the decision the Introduction named. If a question goes unanswered, cut it or say why the data could not answer it.
2. **Titles are claims.** Each slide title, and the first sentence of each Results paragraph, states a finding in a full sentence: "Qwen answered 2.9 points more questions correctly than Llama," not "Accuracy comparison."
3. **Everything shown is used.** Each figure and table supports a claim the Discussion relies on. If it does not, move it to the appendix or drop it.
4. **One name and one number for each thing.** Do not switch among "accuracy," "performance," and "agreement" for the same quantity. Every number matches across the slides, the narration, and the milestones.

5. **The title test.** Read only your slide titles, in order. They should tell the whole argument, recommendation included. If they do not, fix the titles before the slides.

Clients want the answer early, so end the Introduction with a one-sentence preview of your recommendation. The rest of the talk earns it.

Example storyboard

Here is the model choice from Lab 5 told as a story, one slide title per line. Your project will have more methods and results, but the shape is the same.

	Slide title	Shown
I	1. A product team built its tool on Llama 3.1 8B and asks whether to switch to Qwen 2.5 7B, an open model of similar size.	
	2. Switching takes work, so the team will keep Llama if it is no more than 1 point less accurate than Qwen.	Margin, set before seeing the data
	3. Our answer: switch to Qwen, but not for tasks that track objects.	
M	4. Both models answered the same 5,761 multiple-choice questions from 24 BIG-Bench Hard tasks, so we compare them question by question.	Paired table
	5. Questions in a task share a format and a skill, so we resample whole tasks to measure uncertainty.	
R	6. Qwen answered 2.9 points more questions correctly than Llama (53.7% versus 50.8%).	Paired table
	7. The gap varies by task: the 95% interval runs from -0.6 to $+6.6$ points, so Qwen is not shown to be better.	Gap by task
	8. Llama is not shown to be within 1 point of Qwen: it could be up to 6.0 points worse (one-sided 95% bound).	Bootstrap histogram
	9. Qwen does worse than random guessing on all three object-tracking tasks.	Accuracy and chance by task
D	10. Recommendation: switch to Qwen, and route object-tracking questions elsewhere.	
	11. Before launch, test both models on the team's own questions; these benchmark questions are public and may be in the models' training data.	

Read the titles alone and you have the argument. Slides 7 and 8 both say "not shown," and the rule on slide 2 says which one decides. Slide 9 is in Results because the Discussion uses it; a figure no later slide uses would not be.

4. Writing with AI

You may use AI to help write, as on homework. It cannot be your only writer or editor, for two reasons. The argument has to be yours, because you will defend it in the oral without help. And unedited AI prose is hard for a client to use: it is generic where they need specifics, it breaks the argument into bullet fragments, and it uses words like "robust" and "comprehensive" in place of saying what was done.

The rubric grades the writing, not the tool. No criterion asks whether AI wrote something. The criteria ask whether the writing is specific, plain, consistent, and in your team's voice, and unedited AI text usually falls short on those by itself. Four habits are required.

1. **Draft the storyboard yourselves.** On Tue Apr 20, your team writes its slide titles in class without AI. That sequence of claims is your argument. Afterward, use any tools you like to improve it.
2. **Read every sentence aloud** before it goes on a slide or into the video. If you would not say it to the client in those words, rewrite it.

3. **Show one revision.** Milestone 3 includes one passage an AI drafted, next to your revised version, with a sentence or two on what you changed and why.
4. **Narrate in your own words.** Notes are fine. A script read word for word sounds like one.

The words below are not banned. Each one is vague unless you say exactly what it means in your study.

Instead of	Write what you did or found
audit, assess, evaluate	compared, counted, checked, re-ran, read 50 cases
robust	the change you made and what stayed the same: "The recommendation held when we merged the six most-confused breeds."
significant	keep it for a test result, with the p-value or interval; otherwise give the size: "a 2.9-point gap"
comprehensive, holistic, nuanced	delete, or say what was covered
leverage, utilize	use
delve into, explore	look at, compare
key insight, it is important to note, underscores, highlights	state the finding

Bullets are for lists of parallel items, such as data fields or steps. An argument goes in sentences.

5. Deliverables

Due	Deliverable	Points
Thu Mar 4	Team and ranked track choices (Canvas form). Individuals and partial teams may ask to be matched.	—
Thu Mar 18, in class	Launch session: load your track, compute the floor, draft your questions. Not graded.	—
Tue Mar 23	Milestone 1: Introduction and Methods	15
Tue Apr 6	Milestone 2: Results	20
Tue Apr 20	Milestone 3: Discussion and storyboard (completion)	5
Sun Apr 25, 11:59 pm	Final presentation: slides and video. No extensions.	30
Tue Apr 27 or Thu Apr 29, in class	Oral defense	20
Week 15 (date on Canvas)	Peer assessment, one per student	10
	Total	**100**

In-class project work sessions after spring break build pieces of the next milestone. They are not graded separately.

Milestones: one document that grows

The three milestones are one written document. Each milestone adds sections and revises earlier ones. Submit one PDF per team on Canvas with a link to the notebook that reproduces every number. Start each submission with a short list of what changed since the last milestone and why, including how you responded to feedback. Write in paragraphs, and keep code in the notebook. Page limits include figures: Introduction 1 page, Methods 3, Results 4, Discussion 1.

Milestone 1 (Tue Mar 23): Introduction and Methods. Write the Introduction. In Methods, describe the data and what your exploratory analysis found, the floor, the fitted baseline, the AI system, the evaluation

design, and how you will compute uncertainty. Name the assumption you think most threatens your conclusion. End with a paragraph on AI use: what you used it for, what you checked, and one thing it got wrong.

Milestone 2 (Tue Apr 6): Results. Revise the Introduction and Methods. Add the full sensitivity analysis plan to Methods, including the result that would change your recommendation, before you run it. Then write the Results: the fitted baseline, the AI system, the comparison with uncertainty, calibration when the decision uses confidence scores, and where the errors concentrate. Update the AI-use paragraph. Keep the holdout split sealed until this milestone's analysis. Once you have seen holdout results, do not quietly change the main outcome, comparison, subgroup, or sensitivity rule; label later analyses as exploratory.

Milestone 3 (Tue Apr 20): Discussion and storyboard. Add the sensitivity analysis result to the Results and write the Discussion. Attach the storyboard your team wrote in class that day: every slide title in order, grouped by section, with the figure or table each slide shows and who presents it. Include the revision example from Section 4. Rough work is fine; this milestone is graded for completion.

Final presentation: slides and video

Submit the slides as a PDF and the video as instructed on Canvas, by **Sun Apr 25 at 11:59 pm. There are no extensions**, because the instructor watches every video before writing oral questions.

The video is at most five minutes, narrates the slides you submit, and includes all four members speaking. Plan on about 8 to 12 slides: under a minute for the Introduction, about a minute for Methods, about two minutes for Results, and about a minute for the Discussion. After the last presented slide, add appendix slides that are not in the video: details you expect to be asked about in the oral, and one slide on how you used AI in the project (what for, what you checked, what you changed).

Oral defense

Each team meets the instructor in class on Tue Apr 27 or Thu Apr 29 for about five to seven minutes. The instructor has watched your video and read Milestone 3. It is a conversation, not a second presentation, and any member may be asked about any part of the project. Typical questions: Why this baseline? What does your calibration result show? What did the sensitivity analysis change, and what would a different result have meant? Which piece of evidence most supports your recommendation? What would change it? "We don't know, and here is how we would find out" is a good answer.

Peer assessment

Each student privately rates themselves and each teammate on preparation and reliability, contribution to the statistical work, teamwork, and understanding of the whole project, with specific examples. Ratings are not averaged mechanically; the instructor weighs them with the milestones and the oral. If a teammate is not contributing, tell the instructor early rather than waiting for the form.

6. Common questions

What if the baseline beats the AI system? That is a finding, not a failed project.

What if the answer is "we can't tell"? That is a valid conclusion when your uncertainty analysis supports it and you say what data would settle it.

How much code is expected? Less than in a modeling course. The starter notebooks handle data loading and the AI outputs. Your statistical choices and how you explain them are what count.

Can we split the work? Yes, but every member must be able to explain every part. The oral checks this.